

PPU.MAC

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第 1 章

PPU.MAC

```

;=====
; PPU.MAC
; 2006. 2.28
;=====
;copyright = "ppu.mac ver 1.00 Copyright (C) 2006 LdLabo"

; -----
; レジスタ定義
; -----
const R = R0,R1,R2,R3,R4,R5,R6,R7,R8,SR0,SR1,SR2,SR3,SR4,SR5,SR6,FC,FZ,FE

const PC = PC

; -----
; 擬似命令
; -----
macro ORG NUMBER
    $ = #0
endm

macro DS NUMBER
    $ = $+#0
endm

macro DEFS NUMBER
    $ = $+#0
endm

; =====
; 基本命令
; =====

; -----
; NOP
; -----
macro NOP
    db 00,00,00,00
endm

; -----
; JP n'n
; -----
macro JP NUMBER
    db 01,(#0>>10)&0ffh,(#0>>2)&0ffh,0
endm

; -----
; JZ n'n
; -----

```

```
macro      JZ NUMBER
    db 02, (#0>>10)&0ffh, (#0>>2)&0ffh, 0
endm
```

```
; -----
;      JNZ n'n
; -----
```

```
macro      JNZ NUMBER
    db 0ah, (#0>>10)&0ffh, (#0>>2)&0ffh, 0
endm
```

```
; -----
;      LD r0,r1
; -----
```

```
macro      LD R,R
    db 03, #0, #1, 0
endm
```

```
; -----
;      LD r0,n
; -----
```

```
macro      LD R,NUMBER
    db 04, #0, #1&0xff, 0
endm
```

```
; -----
;      AND r0,r1
; -----
```

```
macro      AND R,R
    db 05, #0, #1, 0
endm
```

```
; -----
;      OR r0,r1
; -----
```

```
macro      OR R,R
    db 06, #0, #1, 0
endm
```

```
; -----
;      AND r0,n
; -----
```

```
macro      AND R,NUMBER
    db 07, #0, #1&0xff, 0
endm
```

```
; -----
;      OR r0,n
; -----
```

```
macro      OR R,NUMBER
    db 08, #0, #1&0xff, 0
endm
```

```
; -----
;      USR r0,r1,r2
; -----
```

```
macro      USR R,R,R
    db 09, #0, #1, #2
endm
```

```
; -----
;      LD r0,pc
; -----
```

```
macro      LD R,PC
    db 0bh, #0, 0, 0
endm
```

```

; -----
;   LD pc,r0
; -----
macro      LD PC,R
    db 0ch,#1,0,0
endm

; -----
;   XOR r0,n
; -----
macro      XOR R,NUMBER
    db 0dh,#0,#1&0ffh,0
endm

; -----
;   XOR r0,r1
; -----
macro      XOR R,R
    db 0eh,#0,#1,0
endm

; =====
;   実用命令
; =====

; -----
;   CALL sr0,n'n
; -----
macro      CALL R,NUMBER
    db 0bh,#0,0,0
    db 01,(#1>>10)&0ffh,(#1>>2)&0ffh,0
endm

; -----
;   RET sr0
; -----
macro      RET R
    db 0ch,#0,0,0
endm

; -----
;   INC r0
; -----
macro      INC R
    db 16,#0,0,0
endm

; -----
;   DEC r0
; -----
macro      DEC R
    db 17,#0,0,0
endm

; -----
;   CP r0,r1
; -----
macro      CP R,R
    db 18,#0,#1,0
endm

; -----
;   CP r0,n
; -----
macro      CP R,NUMBER

```

```

    db 19,#0,#1&0ffh,0
endm

```

```

; -----
;      ADD r0,r1
; -----
macro    ADD R,R
    db 20,#0,#1,0
endm

```

```

; -----
;      ADD r0,n
; -----
macro    ADD R,NUMBER
    db 21,#0,#1&0ffh,0
endm

```

```

; -----
;      DIV r0,r1
; -----
macro    DIV R,R
    db 22,#0,#1,0
endm

```

```

; -----
;      DIV r0,n
; -----
macro    DIV R,NUMBER
    db 23,#0,#1&0ffh,0
endm

```

```

; -----
;      SUB r0,r1
; -----
macro    SUB R,R
    db 24,#0,#1,0
endm

```

```

; -----
;      SUB r0,n
; -----
macro    SUB R,NUMBER
    db 25,#0,#1&0ffh,0
endm

```

```

; -----
;      SLA r0,r1
; -----
macro    SLA R,R
    db 26,#0,#1,0
endm

```

```

; -----
;      SLA r0,n
; -----
macro    SLA R,NUMBER
    db 27,#0,#1&0ffh,0
endm

```

```

; -----
;      SRL r0,r1
; -----
macro    SRL R,R
    db 28,#0,#1,0
endm

```

```
; -----  
;      SRL r0,n  
; -----  
macro      SRL R,NUMBER  
    db 29,#0,#1&0ffh,0  
endm
```